

Student Lab Manual - Working with Linux Containers using Docker

Lab Objective

Learn how to pull a Linux image, start a Linux container, work with the operating system, manage files, users, networking and packages.

Exercise 1 - Pull Ubuntu Image

```
docker pull ubuntu:24.04
```

docker images

```
Last login: Fri Jun 19 11:40:20 on ttys000
/dev/fd/12:25: command not found: compdef
vaibhavgupta@FVF098WQ05F-vaibhavgupta Day 41 June 22 % docker pull ubuntu:24.04

docker images

24.04: Pulling from library/ubuntu
532a4cf7e47b: Download complete
Digest: sha256:786a8b558f7be160c6c8c4a54f9a57274f3b4fb1491cf65146521ae77ff1dc54
Status: Downloaded newer image for ubuntu:24.04
docker.io/library/ubuntu:24.04

What's next:
View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu:24.04

IMAGE          ID              DISK USAGE    CONTENT SIZE  EXTRA
alpine:latest  28bd5fe8b56d    13.6MB        4.27MB        U
compose-postgres-api-lab-api:latest  aebc9580f8bf    387MB         97MB          U
customer-api:1.0  62593972ba26    360MB         88.3MB        U
dmgw/padmin4:latest  cefc4ccab7c9    765MB         172MB         U
node:22-alpine  e58326d8d441    229MB         58.5MB         U
postgres:16     881f1bc7bd5e    663MB         165MB         U
postgres:latest  29ee7bb38d98    671MB         167MB         U
ubuntu:24.04    786a8b558f7b    141MB         38.8MB         U
weather-ui:1.0  399165a2f368    76.1MB        21.9MB         U
weatherapi:1.0  aefb12c8db89    351MB         88.4MB         U
weatherapi:single  8cf0eb18e30e    1.25GB        311MB          U
vaibhavgupta@FVF098WQ05F-vaibhavgupta Day 41 June 22 %
```

Exercise 2 - Start Ubuntu Container

```
docker run -it ubuntu:24.04 bash
```

```
vaibhavgupta@FVF098WQ05F-vaibhavgupta Day 41 June 22 % docker run -it ubuntu:24.04 bash
root@13879ad22f98:/# pwd
/
```

Exercise 3 - Linux Navigation

```
pwd
```

```
ls
```

```
ls -la
```

```
cd /etc
```

```
ls
```

```
root@13079ad22f80:/# pwd
/
root@13079ad22f80:/# ls
bin boot dev etc home lib media mnt opt proc root run sbin srv sys usr var
root@13079ad22f80:/# ls -la
total 56
drwxr-xr-x 1 root root 4096 Jun 22 04:25 .
drwxr-xr-x 1 root root 4096 Jun 22 04:25 ..
-rwxr-xr-x 1 root root 0 Jun 22 04:25 .dockerenv
lrwxrwxrwx 1 root root 7 Apr 22 2024 bin -> usr/bin
drwxr-xr-x 2 root root 4096 Apr 22 2024 boot
drwxr-xr-x 5 root root 360 Jun 22 04:25 dev
drwxr-xr-x 1 root root 4096 Jun 22 04:25 etc
drwxr-xr-x 3 root root 4096 May 9 14:23 home
lrwxrwxrwx 1 root root 7 Apr 22 2024 lib -> usr/lib
drwxr-xr-x 2 root root 4096 May 9 14:17 media
drwxr-xr-x 2 root root 4096 May 9 14:17 mnt
drwxr-xr-x 2 root root 4096 May 9 14:17 opt
dr-xr-xr-x 217 root root 0 Jun 22 04:25 proc
drwxr-xr-x 2 root root 4096 May 9 14:23 root
drwxr-xr-x 4 root root 4096 May 9 14:23 run
lrwxrwxrwx 1 root root 0 Apr 22 2024 sbin -> usr/sbin
drwxr-xr-x 2 root root 4096 May 9 14:17 srv
dr-xr-xr-x 11 root root 0 Jun 22 04:25 sys
drwxrwxrwt 2 root root 4096 May 9 14:23 tmp
drwxr-xr-x 11 root root 4096 May 9 14:17 var
drwxr-xr-x 11 root root 4096 May 9 14:23 vvar
root@13079ad22f80:/# cd /etc
root@13079ad22f80:/etc# ls
alternatives cron.d dpkg gnutls host.conf issue.net legal machine-id nsswitch.conf passwd rc1.d rc6.d selinux subgid sysctl.d
apt cron.daily e2scrub.conf group hostname kernel libaudit.conf mke2fs.conf opt os-release profile rc3.d rc5.d shadow subuid systemd
bash.bashrc debconf.conf environment group- hosts ld.so.cache login.defs mtab pam.conf profile rc4.d rmt shadow- subuid- terminfo
bindresvport.blacklist debian_version fstab gshadow- init.d ld.so.conf logrotate.d networks pam.d rc0.d rc5.d security skel sysctl.conf xattr.conf
cloud default gai.conf gshadow- issue ld.so.conf.d lsb-release notes.txt passw rc1.d rc6.d security skel sysctl.conf xattr.conf
root@13079ad22f80:/etc#
```

Exercise 4 - Operating System Information

cat /etc/os-release

whoami

```
root@13079ad22f80:/etc# cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
root@13079ad22f80:/etc# whoami
root
root@13079ad22f80:/etc#
```

Exercise 5 - File Management

touch notes.txt

echo "Hello Docker" > notes.txt

cat notes.txt

```
root@13079ad22f80:/etc# touch notes.txt
root@13079ad22f80:/etc# echo "Hello Docker" > notes.txt
root@13079ad22f80:/etc# cat notes.txt
Hello Docker
root@13079ad22f80:/etc# ls
alternatives cron.d dpkg gnutls host.conf issue.net legal machine-id nsswitch.conf passwd rc1.d rc6.d selinux subgid sysctl.d
apt cron.daily e2scrub.conf group hostname kernel libaudit.conf mke2fs.conf opt os-release profile rc3.d rc5.d shadow- subuid- terminfo
bash.bashrc debconf.conf environment group- hosts ld.so.cache login.defs mtab pam.conf profile rc4.d rmt shadow- subuid- terminfo
bindresvport.blacklist debian_version fstab gshadow- init.d ld.so.conf logrotate.d networks pam.d rc0.d rc5.d security skel sysctl.conf xattr.conf
cloud default gai.conf gshadow- issue ld.so.conf.d lsb-release notes.txt passw rc1.d rc6.d security skel sysctl.conf xattr.conf
root@13079ad22f80:/etc#
```

Exercise 6 - Directories

mkdir training

cd training

touch test.txt

```
root@13079ad22f80:/etc# mkdir training
root@13079ad22f80:/etc# ls
alternatives          environment          gshadow             issue              legal              mke2fs.conf        os-release          profile.d          rcS.d              selinux            subgid             terminfo
apt                  debconf.conf        fstab              gshadow-          issue.net          libaudit.conf      mtab               pam.conf          rc0.d             shadow            subuid             training
bash.bashrc         debian_version      gai.conf           host.conf         kernel            login.defs         networks           pam.d            rc1.d             shadow-          subuid-          update-motd.d
bindresvport.blacklist default             gnutils           hostname          ld.so.cache       logrotate.d        notes.txt          passwd            rc2.d             resolv.conf       shells            sysctl.conf      xattr.conf
cloud               dhcpcd              group             hosts            ld.so.conf        lib-release        nsswitch.conf      passwd-          rc3.d             rmt              shell            sysctl.d
cron.d              e2scrub.conf       group-            init.d            ld.so.conf.d      machine-id         opt               profile          rc4.d             security          subgid           systemd
```

```
root@13079ad22f80:/etc# cd training
root@13079ad22f80:/etc/training# touch test.txt
root@13079ad22f80:/etc/training# echo "Hello I am vaibhav" > test.txt
root@13079ad22f80:/etc/training# cat test.txt
Hello I am vaibhav
root@13079ad22f80:/etc/training# ls
test.txt
root@13079ad22f80:/etc/training#
```

Exercise 7 - Processes

ps -ef

```
test.txt
root@13079ad22f80:/etc/training# ps -ef
UID          PID  PPID  C  STIME TTY          TIME CMD
root           1      0  0 04:25 pts/0    00:00:00 bash
root          22      1  3 04:36 pts/0    00:00:00 ps -ef
root@13079ad22f80:/etc/training#
```

Exercise 8 - Install Software

apt update

apt install curl -y

curl -version

```
root@13079ad22f80:/etc/training# apt update
Get:1 http://ports.ubuntu.com/ubuntu-ports noble InRelease [256 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports noble-updates InRelease [126 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-backports InRelease [126 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports noble-security InRelease [126 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports noble/restricted arm64 Packages [113 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 Packages [1776 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 Packages [1297 kB]
Get:8 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 Packages [19.0 MB]
Get:9 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted arm64 Packages [2020 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports noble-updates/multiverse arm64 Packages [63.8 kB]
Get:11 http://ports.ubuntu.com/ubuntu-ports noble-backports/main arm64 Packages [36.3 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe arm64 Packages [2898 kB]
Get:13 http://ports.ubuntu.com/ubuntu-ports noble-backports/multiverse arm64 Packages [671 B]
Get:14 http://ports.ubuntu.com/ubuntu-ports noble-backports/main arm64 Packages [48.9 kB]
Get:15 http://ports.ubuntu.com/ubuntu-ports noble-security/main arm64 Packages [58.3 kB]
Get:16 http://ports.ubuntu.com/ubuntu-ports noble-security/restricted arm64 Packages [1933 kB]
Get:17 http://ports.ubuntu.com/ubuntu-ports noble-security/universe arm64 Packages [1519 kB]
Get:18 http://ports.ubuntu.com/ubuntu-ports noble-security/multiverse arm64 Packages [59.5 kB]
Get:19 http://ports.ubuntu.com/ubuntu-ports noble-security/main arm64 Packages [995 kB]
Fetched 31.9 MB in 5s (6158 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
6 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@13079ad22f80:/etc/training# apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ca-certificates krb5-locales libbrotli1 libcurl4t64 libgssapi-krb5-2 libk5crypto3 libkeyutils1 libkrb5-3 libkrb5support0 libldap-common libldap2 libnghttp2-14 libpsl5t64 librtmp1 libssl2-2
  libssl2-modules libssl2-modules-db libssh-4 libssl3t64 openssl publicsuffix
Suggested packages:
  krb5-doc krb5-user libssl2-modules-gssapi-mit | libssl2-modules-gssapi-heimdal libssl2-modules-ldap libssl2-modules-otp libssl2-modules-sql
The following NEW packages will be installed:
  ca-certificates curl krb5-locales libbrotli1 libcurl4t64 libgssapi-krb5-2 libk5crypto3 libkeyutils1 libkrb5-3 libkrb5support0 libldap-common libldap2 libnghttp2-14 libpsl5t64 librtmp1 libssl2-2
  libssl2-modules libssl2-modules-db libssh-4 openssl publicsuffix
The following packages will be upgraded:
  libssl3t64
1 upgraded, 21 newly installed, 0 to remove and 5 not upgraded.
Need to get 5339 kB of archives.
After this operation, 9848 kB of additional disk space will be used.
Get:1 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libssl3t64 arm64 3.0.13-0ubuntu3.11 [1799 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 openssl arm64 3.0.13-0ubuntu3.11 [906 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 ca-certificates all 20240409~24.04.1 [139 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 krb5-locales all 1.20.1-0ubuntu2.6 [14.8 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libkrb5support0 arm64 1.20.1-0ubuntu2.6 [34.8 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libk5crypto3 arm64 1.20.1-0ubuntu2.6 [86.8 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libkeyutils1 arm64 1.4.3-3build1 [9454 B]
Get:8 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libkrb5-3 arm64 1.20.1-0ubuntu2.6 [349 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libgssapi-krb5-2 arm64 1.20.1-0ubuntu2.6 [142 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libnghttp2-14 arm64 1.59.0-ubuntu0.3 [74.6 kB]
Get:11 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libpsl5t64 arm64 0.21.2-1.1build1 [57.4 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 publicsuffix all 20231001.0357-0.1 [129 kB]
Get:13 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libbrotli1 arm64 1.1.0-2build2 [339 kB]
Get:14 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libssl2-modules-db arm64 2.1.28dfsg1-subuntu0.1 [21.6 kB]
Get:15 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libssl2-2 arm64 2.1.28dfsg1-subuntu0.1 [54.7 kB]
Get:16 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libldap2 arm64 2.6.10+dfsg-0ubuntu0.24.04.1 [196 kB]
```

```
Unpacking libldap2:arm64 (2.6.10+dfsg-0ubuntu0.24.04.1) ...
Selecting previously unselected package librtmp1:arm64.
Preparing to unpack .../15-librtmp1_2.4+20151223.gitfa864ed.1-2build7_arm64.deb ...
Unpacking librtmp1:arm64 (2.4+20151223.gitfa864ed.1-2build7) ...
Selecting previously unselected package libssh-4:arm64.
Preparing to unpack .../16-libssh-4_0.10.6-2ubuntu0.4_arm64.deb ...
Unpacking libssh-4:arm64 (0.10.6-2ubuntu0.4) ...
Selecting previously unselected package libcurl4t64:arm64.
Preparing to unpack .../17-libcurl4t64_8.5.0-2ubuntu10.9_arm64.deb ...
Unpacking libcurl4t64:arm64 (8.5.0-2ubuntu10.9) ...
Selecting previously unselected package curl.
Preparing to unpack .../18-curl_8.5.0-2ubuntu10.9_arm64.deb ...
Unpacking curl (8.5.0-2ubuntu10.9) ...
Selecting previously unselected package libldap-common.
Preparing to unpack .../19-libldap-common_2.6.10+dfsg-0ubuntu0.24.04.1_all.deb ...
Unpacking libldap-common (2.6.10+dfsg-0ubuntu0.24.04.1) ...
Selecting previously unselected package libsasl2-modules:arm64.
Preparing to unpack .../20-libsasl2-modules_2.1.28+dfsg1-5ubuntu3.1_arm64.deb ...
Unpacking libsasl2-modules:arm64 (2.1.28+dfsg1-5ubuntu3.1) ...
Setting up libkeyutils:arm64 (1.6.2-3build1) ...
Setting up libbrotli1:arm64 (1.1.0-2build2) ...
Setting up libsasl2-modules:arm64 (2.1.28+dfsg1-5ubuntu3.1) ...
Setting up libpsl5t64:arm64 (0.21.2-1.1build1) ...
Setting up libnghttp2-14:arm64 (1.59.0-2ubuntu0.3) ...
Setting up krb5-locales (1.20.1-6ubuntu2.6) ...
Setting up libldap-common (2.6.10+dfsg-0ubuntu0.24.04.1) ...
Setting up libkrb5support0:arm64 (1.20.1-6ubuntu2.6) ...
Setting up libsasl2-modules-db:arm64 (2.1.28+dfsg1-5ubuntu3.1) ...
Setting up librtmp1:arm64 (2.4+20151223.gitfa864ed.1-2build7) ...
Setting up libk5crypto3:arm64 (1.20.1-6ubuntu2.6) ...
Setting up libsasl2-2:arm64 (2.1.28+dfsg1-5ubuntu3.1) ...
Setting up libkrb5-3:arm64 (1.20.1-6ubuntu2.6) ...
Setting up openssl (3.0.13-0ubuntu3.11) ...
Setting up publicsuffix (20231001.0357-0.1) ...
Setting up libldap2:arm64 (2.6.10+dfsg-0ubuntu0.24.04.1) ...
Setting up ca-certificates (20240601-24.04.1) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/aarch64-linux-gnu/perl/5.38.2 /usr/local/share/perl/5.38.2 /usr/lib/aarch64-linux-gnu/perl-base /usr/lib/aarch64-linux-gnu/perl/5.38 /usr/share/perl/5.38 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8.)
debconf: falling back to frontend: Teletype
Updating certificates in /etc/ssl/certs...
121 added, 0 removed; done.
Setting up libgssapi-krb5-2:arm64 (1.20.1-6ubuntu2.6) ...
Setting up libssh-4:arm64 (0.10.6-2ubuntu0.4) ...
Setting up libcurl4t64:arm64 (8.5.0-2ubuntu10.9) ...
Setting up curl (8.5.0-2ubuntu10.9) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for ca-certificates (20240601-24.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@13079ad22f80:/etc/training#
```

```
root@13079ad22f80:/etc/training# curl -s34212001223version
curl: (3) URL using bad/illegal format or missing URL
root@13079ad22f80:/etc/training# curl version
curl: (6) Could not resolve host: version
root@13079ad22f80:/etc/training# curl --version
curl 8.5.0 (aarch64-unknown-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.2 (+libidn2/2.3.7) libssh/0.10.6/openssl/zlib nghttp/1.59.0 librtmp/2.3 Op
enLDAP/2.6.18
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.9
Protocols: dict file ftp gopher gophers http https imap ldaps mqtt pop3 pop3s rtsp scp sftp smb smbs smtp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
root@13079ad22f80:/etc/training#
```

Exercise 9 - Networking

hostname

hostname -I

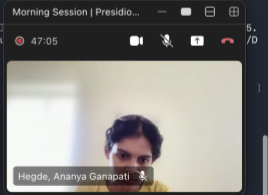
apt install iputils-ping -y

ping google.com


```

Features: alt-svc AsyncDNS brotli GSS-API IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
root@13079ad22f80:/etc/training# hostname
13079ad22f80
root@13079ad22f80:/etc/training# hostname -I
172.17.0.2
root@13079ad22f80:/etc/training# apt install iputils-ping -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcap2-bin libpam-cap
The following NEW packages will be installed:
  iputils-ping libcap2-bin libpam-cap
0 upgraded, 3 newly installed, 0 to remove and 5 not upgraded.
Need to get 90.6 kB of archives.
After this operation, 598 kB of additional disk space will be used.
Get:1 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libcap2-bin arm64 1:2.66-Subuntu2.4 [33.5 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 iputils-ping arm64 3:20240117-1ubuntu0.1 [44.6 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libpam-cap arm64 1:2.66-Subuntu2.4 [12.5 kB]
Fetched 90.6 kB in 1s (171 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libcap2-bin.
(Reading database ... 4984 files and directories currently installed.)
Preparing to unpack .../libcap2-bin_3:20240117-1ubuntu0.1_arm64.deb ...
Unpacking libcap2-bin (1:2.66-Subuntu2.4) ...
Selecting previously unselected package iputils-ping.
Preparing to unpack .../iputils-ping_3:20240117-1ubuntu0.1_arm64.deb ...
Unpacking iputils-ping (3:20240117-1ubuntu0.1) ...
Selecting previously unselected package libpam-cap:arm64.
Preparing to unpack .../libpam-cap_1:2.66-Subuntu2.4_arm64.deb ...
Unpacking libpam-cap:arm64 (1:2.66-Subuntu2.4) ...
Setting up libcap2-bin (1:2.66-Subuntu2.4) ...
Setting up libpam-cap:arm64 (1:2.66-Subuntu2.4) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/aarch64-linux-gnu/perl/5.38 /usr/share/perl5 /usr/lib/aarch64-linux-gnu/perl-base /usr/lib/aarch64-linux-gnu/perl/5.38 /usr/share/perl/5.38 /usr/lib/aarch64-linux-gnu/perl-modules/5.38))
debconf: falling back to frontend: Teletype
Setting up iputils-ping (3:20240117-1ubuntu0.1) ...
root@13079ad22f80:/etc/training# ping google.com
PING google.com (142.251.221.101) 64(84) bytes of data.
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=1 ttl=63 time=33.7 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=2 ttl=63 time=30.3 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=3 ttl=63 time=36.0 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=4 ttl=63 time=25.8 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=5 ttl=63 time=23.0 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=6 ttl=63 time=43.0 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=7 ttl=63 time=24.9 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=8 ttl=63 time=25.7 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=9 ttl=63 time=33.1 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=10 ttl=63 time=36.4 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=11 ttl=63 time=24.7 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=12 ttl=63 time=36.7 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=13 ttl=63 time=29.4 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=14 ttl=63 time=34.9 ms

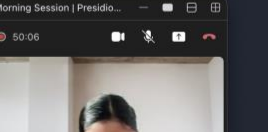
```



```

64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=152 ttl=63 time=24.1 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=153 ttl=63 time=29.2 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=154 ttl=63 time=45.3 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=155 ttl=63 time=88.7 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=156 ttl=63 time=81.6 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=157 ttl=63 time=290 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=158 ttl=63 time=32.1 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=159 ttl=63 time=34.2 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=160 ttl=63 time=31.7 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=161 ttl=63 time=42.6 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=162 ttl=63 time=33.4 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=163 ttl=63 time=28.6 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=164 ttl=63 time=29.7 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=165 ttl=63 time=27.2 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=166 ttl=63 time=40.2 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=167 ttl=63 time=34.8 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=168 ttl=63 time=28.9 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=169 ttl=63 time=29.3 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=170 ttl=63 time=30.6 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=171 ttl=63 time=22.8 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=172 ttl=63 time=22.8 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=173 ttl=63 time=29.0 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=174 ttl=63 time=29.8 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=175 ttl=63 time=42.6 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=176 ttl=63 time=30.1 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=177 ttl=63 time=36.9 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=178 ttl=63 time=25.3 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=179 ttl=63 time=27.9 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=180 ttl=63 time=53.1 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=181 ttl=63 time=31.6 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=182 ttl=63 time=51.1 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=183 ttl=63 time=35.9 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=184 ttl=63 time=34.9 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=185 ttl=63 time=24.0 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=186 ttl=63 time=41.6 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=187 ttl=63 time=30.7 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=188 ttl=63 time=33.2 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=189 ttl=63 time=26.8 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=190 ttl=63 time=106 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=191 ttl=63 time=90.5 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=192 ttl=63 time=147 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=193 ttl=63 time=37.3 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=194 ttl=63 time=49.5 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=195 ttl=63 time=179 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=196 ttl=63 time=81.6 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=197 ttl=63 time=126 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=198 ttl=63 time=25.3 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=199 ttl=63 time=31.5 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=200 ttl=63 time=29.3 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=201 ttl=63 time=35.1 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=202 ttl=63 time=24.6 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=203 ttl=63 time=24.9 ms
64 bytes from cgk03a03-in-f14.1e100.net (142.251.221.110): icmp_seq=204 ttl=63 time=34.4 ms
^C
  google.com ping statistics
204 packets transmitted, 204 received, 0% packet loss, time 204109ms
rtt min/avg/max/mdev = 19.426/46.996/290.314/39.867 ms
root@13079ad22f80:/etc/training#

```



Exercise 10 - Users

useradd student

grep student /etc/passwd

```
root@13b79ad22f80:/etc/trainings touch secret.txt
root@13b79ad22f80:/etc/trainings chmod 777 secret.txt
root@13b79ad22f80:/etc/trainings ls -l
total 4
-rwxrwxrwx 1 root root 0 Jun 22 04:47 secret.txt
-rw-r--r-- 1 root root 19 Jun 22 04:43 test.txt
root@13b79ad22f80:/etc/trainings
```

[illegible]

```

root@13b7ad22f80:/etc/trainings exit
exit
vaibhavgupta@VFVG98WQ8F-vaibhavgupta Day 41 June 22 of docker ps -a
CONTAINER ID        IMAGE               COMMAND              CREATED              STATUS              PORTS              NAMES
13b7ad22f80        ubuntu:24.04       "bash"              24 minutes ago      Exited (0) 12 seconds ago              priceless_gagarin
09e5a352b86a       compose-postgres-  "dotnet CustomerApi..." 3 days ago          Exited (0) 3 days ago              compose-customer-api
3ab6a64fb793       postgres:16        "docker-entrypoint.s..." 3 days ago          Exited (0) 3 days ago              compose-postgres-db
bfff6382e358       postgres:16        "docker-entrypoint.s..." 4 days ago          Exited (0) 3 days ago              compose-lab-db-1
929e410e4141latest  "pgadmin4"         "pgadmin4"           4 days ago          Exited (137) 3 days ago              compose-lab-pgadmin-1
c5938263743a       customer-api:1.1.0 "dotnet CustomerApi..." 4 days ago          Exited (0) 4 days ago              customer-api
129348437874       postgres:16        "docker-entrypoint.s..." 5 days ago          Exited (0) 4 days ago              postgres-db
d3964c56a347       postgres:latest    "docker-entrypoint.s..." 6 days ago          Exited (1) 6 days ago              dreamy_solomon
d97851c1efb0       postgres           "docker-entrypoint.s..." 6 days ago          Exited (0) 6 days ago              some-postgres
vaibhavgupta@VFVG98WQ8F-vaibhavgupta Day 41 June 22 of docker start 13b7ad22f80
13b7ad22f80
vaibhavgupta@VFVG98WQ8F-vaibhavgupta Day 41 June 22 of docker attach 13b7ad22f80
root@13b7ad22f80:/#

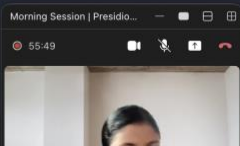
```

```
docker run -it alpine sh
```

Vaibhav Gupta vgupta1@presidio.com

```
8:11
vaibhavgupta@FVF0985MQ85F-vaibhavgupta Day 41 June 22 % docker pull alpine
Using default tag: latest
latest: Pulling from library/alpine
Digest: sha256:28bd5fe8b56d1b0048e5babf5b10710eebe8bae67db86916198adeec434943f8b
Status: Image is up to date for alpine:latest
docker.io/library/alpine:latest

What's next:
View a summary of image vulnerabilities and recommendations → docker scout quickview alpine
vaibhavgupta@FVF0985MQ85F-vaibhavgupta Day 41 June 22 % docker run -it alpine sh
/ #
```



Knowledge Check

What is an image? What is a container? What command creates a file? What command installs software?